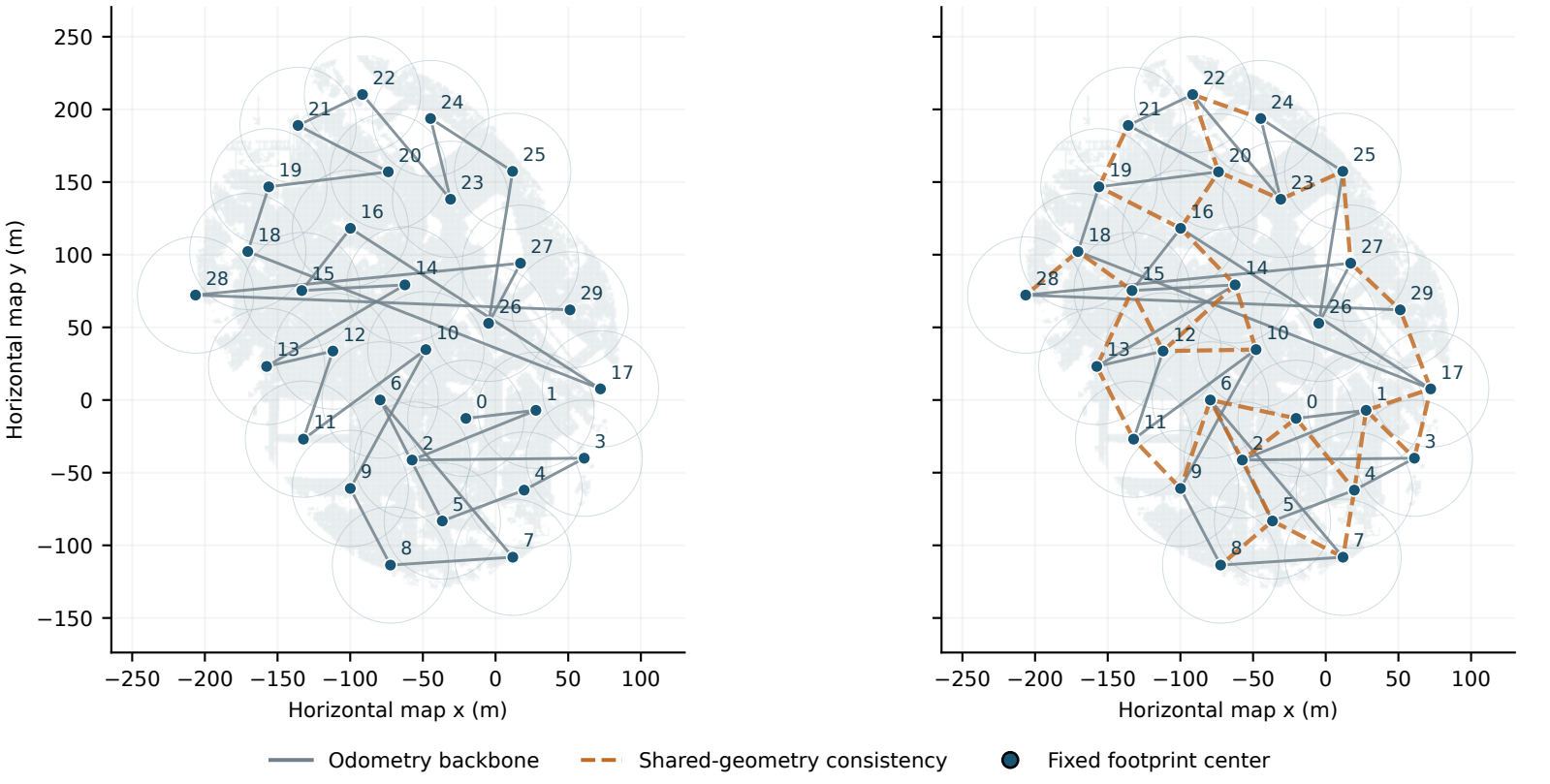

(a) Chronological backbone
30 nodes · 29 edges · no local cycles
(b) Sparse spatial graph
29 backbone + 33 consistency edges · 33 cycles
Horizontal map y (m)
Horizontal map x (m)
Horizontal map x (m)
Odometry backbone
Shared-geometry consistency
Fixed footprint center
A05 · R = 40 m · at most 3 spatial neighbors per node · inter-robot loops omitted
Graph layout uses footprint centers; optimized variables retain their original creation-anchor IMU frames.